

Sentiment analysis of player chat messaging in the video game StarCraft 2: Extending a lexicon-based model



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ABSTRACT

There is a growing need for automated tools which make predictions about the positivity or negativity of sentiment conveyed by text. Such tools have a number of important applications in game user research. They are useful for understanding users generally, as they may give Big Data researchers access to a new source of information about player learning environments. Sentiment analysis methods are also applicable to the detection of toxicity, and the identification of players or player messages that are a potential threat to the player experience. A major challenge in sentiment analysis, however, is developing portable models that can be applied to new domains with relatively little effort. In the present study we extend a lexicon-based sentiment extractor, SO-CAL, to the analysis of instant messages across 1000 games of StarCraft 2. We show that, with updates to dictionary entries that are tailored to the classification task at hand, SO-CAL constitutes a respectable classifier of sentiment and toxicity that is robust across differences in player region and league. We verify the performance of our toxicity detector against a sample of 2025 additional games. Our results support the proposal that lexicon-based sentiment extraction is a useful and portable method of sentiment analysis, and that it can be deployed to identify toxicity.

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1. Introduction

1.1. Sentiment analysis and game commentary

Sentiment Analysis (also known as opinion mining) is the process of extracting select information from a text and determining the attitude that the writer is trying to express through language (see reviews in [1,2]). Sentiment extractors in their most basic form categorize texts as either having a positive or negative (or sometimes neutral) sentiment. A common application of sentiment analysis is the automatic determination of whether an online review (of a movie, a book, or a consumer product) contains a positive or negative assessment. A more fine-grained sentiment analysis task may involve differentiating multiple sentiments embedded within one text. For instance, a sophisticated sentiment analysis program may be able to extract from a restaurant review that a critic had a positive opinion about the food, but a negative opinion about the service.

With the growing popularity of big data, sentiment analysis is now a widespread tool in the repertoire of social media analysis carried out by companies rolling out new products, by political parties and campaign managers, and by customer service representatives attempting to assess the overall opinion of customers towards a company. More general applications try to gauge the 'mood of the web', such as the Hedonometer project [3], which measures the level of happiness expressed in Twitter, by aggregating the words and expressions in individual tweets. Once in place, sentiment extractors can serve as important stepping stones for future Big Data science. Cody et al. [4], for example, used the Hedonometer algorithms to study how sentiment on Twitter was influenced by climate change news. Similarly, Fan et al. [5] used emotion classification algorithms to examine the propagation of joy, anger, and sadness through social media.

The rise of online multiplayer games has created further avenues for sentiment analysis research. Online games leave digital footprints of both in-game performance and interplayer communication. One application of automated sentiment analysis in games user research lies in understanding the social context in which learning occurs. While the introduction of big data methods has allowed researchers to begin measuring performance directly over thousands of hours of play [6–9], it remains difficult to derive any

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measures of the social context during this learning. Studying learning in a complex and competitive environment, without also having access to measures of social context, is problematic in light of dominant theory, which suggests that social constraints are crucial in expert development. These include not only factors pertaining to domain knowledge (such as knowledge of technique), but also motivational constraints which impact practice frequency and quality [10]. If sentiment can be effectively identified in the chat history of video game play, future research could examine how the presence of positive and negative evaluations in the learning environment relate to practice frequency, practice quality, and longer term learning.

Another emerging application of sentiment analysis in online games lies in the identification of toxicity - the intentional use of abusive, offensive language, often involving personal attacks [11–13]. Extremely negative sentiment is a threat to the enjoyment of the game community and thus the longevity of the game. There is evidence that toxicity can drive players from video game sessions or lead them to quit playing a game permanently [14]. From the standpoint of game companies [15], or anyone who oversees an online community [16], the capacity to automatically detect toxicity and efficiently dissuade these behaviours, through the placement of bans or other punishments on consistently toxic players, could be integral to a community's survival, and a company's financial success.

Finally, sentiment classification and toxicity detection are very similar tasks so it makes sense to pursue them concurrently within the same work. When toxicity is operationally defined as extremely negative sentiment, the two tasks fall under the same umbrella of sentiment analysis. Furthermore, by classifying sentiment and detecting toxicity together, we can better understand how the two tasks are methodologically distinct.

1.2. Approaches to sentiment analysis

The task in sentiment analysis consists of developing a procedure such that, when a new material needs to be processed, be it a tweet, sentence, headline, excerpt or whole text, its characteristics can be used to decide whether it contains positive or negative sentiment. Two main approaches exist to the problem: machine learning and lexicon-based sentiment extractors. In the machine learning approach, a classifier is trained on a set of pre-labelled instances of other items (sentences, documents, etc.). The classifier *learns* which characteristics distinguish a positive from a negative text. The characteristics which models are sensitive to tend to be the presence of individual words or tokens that are more frequent in the training dataset.

The advantages of the machine learning approach are that, given a labelled dataset, that is, one where documents have been previously determined to be positive or negative, training is trivial, and a classifier can be built quite quickly with existing tools (e.g. [17]). For instance, a number of classifiers have been built using a set of 2000 movie reviews, labelled according to whether their evaluation of the movie discussed is positive or negative [18]. Their performance in most cases is around 80% or above. That is, the resulting classifiers are able to correctly determine the polarity of novel data 80% of the time ([19–24], among many others). Indeed, these approaches have been effective in a variety of domains [25–28].

While machine learning approaches are desirable thanks to their accuracy, they often suffer from a number of disadvantages. First of all, because they are trained on labelled data, they are typically not portable to new types of data. Applying the model to new contexts and datasets will typically require new training data and thus extensive human coding. Models trained on movie reviews will suffer from bias towards that data, and cannot be expected

to be able to capture some of the nuances and particular characteristics in new domains, such as in-game messaging. Building a new classifier would require new training examples, necessitating extensive human coding of game commentary.

Another disadvantage of many machine learning methods is that they lack transparency. While it is possible to examine the features that the classifier is using, these are often not made available, and thus the classification remains in a black box. This is especially the case with neural net classifiers [30–32]. In the case of assessing toxicity in online games, a lack of transparency could pose a problem. Because there are negative repercussions for being labelled as toxic in online communities, such as the suspension or banning of accounts [29], it becomes important that toxicity detectors be transparent and their classifications justifiable. Transparency is especially important in ensuring that the algorithm is fair, and does not unknowingly discriminate, for example, against a particular ethnicity or religion. Justifiability is important when the results might lead to accusations of toxicity or punishments in response to it.

The other main approach to sentiment analysis is the lexicon-based method. On this approach, sentiment scores of text are derived from the sentiment scores of the individual words in the text. The sentiment scores provided by the model's dictionary reflect a word's polarity (e.g., *excellent* is positive; *horrible* is negative). When a new text is encountered, words in the text are matched to words in the dictionary and, using various algorithms, their values aggregated into a sentiment score for the entire text.

Various methods exist to create the dictionary, from manual to automatic, including the use of machine learning methods to automatically generate a dictionary. In either case, lexicon-based methods have been shown to be robust across different domains without changing the dictionaries [33]. In cases where the domain is very specific, a custom dictionary can be created, as Loughran and McDonald [34] did for the financial domain. Furthermore, Brooke et al. [35] also showed that porting dictionaries to a new language or a new domain is not an onerous task, and probably one less onerous than labelling data in a new domain to be used to train a classifier.

Once a reliable dictionary is available, the next task in a lexicon-based method consists of deciding how to aggregate the words found in the text. The simplest approach is to average all words. A text with ten positive and two negative words would then be labelled as positive. Research has shown, however, that accuracy can be significantly improved by taking into account the role of valence shifters [36,37], which include negatives ("not"), intensifiers ("very"), and downtoners ("kind of").

Analysis is further improved when taking into account full phrases, rather than individual words, and by assigning weights to sentences according to where they appear in the text, with sentences towards the end of the text having higher weights, because that is where evaluation is typically summarized [36,38]. Another way of weighting text consists of identifying which parts consist of evaluation, and which parts are mostly description. In reviews in particular, there may be description of the product or of the context that is irrelevant to the evaluation. A movie review might, for instance, describe the plot and the actors' previous roles, both of which are irrelevant to the evaluation of the movie itself. Taboada et al. [39], proposed a method to automatically classify paragraphs in the text as description or evaluation, and showed that it improved the accuracy of the sentiment analysis.

Regardless of the specific rules employed, however, the primary advantage of lexicon-based approaches is their portability. When the model is applied to new domains, researchers do not need to retrain the model. They need only update dictionaries to better represent the new domain. Where the sentiment of individual words does not shift radically, a lexicon-based approach can also

be cumulative across domains and datasets. For example, for every improvement to the model's dictionary, one might rerun a model on *all* archived data to see how prior classifications change. Of course, the practical utility of this approach remains an open empirical question. Specifically, it depends on the performance costs of generalizing lexicon-based models to new datasets, and on how extensively the model must be modified in order to accommodate new datasets. That is, it depends upon the nature of between-domain differences in how sentiment is carried in chat.

Sentiment analysis is an active area of research, and in particular machine learning methods have significantly changed with the explosion of deep learning methods [30,40,41]. All the recent progress notwithstanding, sentiment analysis is not a solved problem, and we are interested in testing the limits of existing methods, and their portability. That is why we deploy a lexicon-based method to the problem of detecting toxicity in online video game chat.

1.3. Applying sentiment analysis to online video game chat

The research question we address in this paper is how well lexicon-based methods for sentiment analysis translate to newer genres of social media, in particular, online video game chat. Recent sentiment analysis research in the domain of tweets, blogs and Facebook posts has shown that adaptations are always necessary. Davidov et al. [42] added punctuations, capitalized letters, and emoticons as features in their classifier, as did Barbosa and Feng [43]. Huang et al. [44] used emoticons as seeds to generate their lexicon. One article studying text from MySpace [45] modified an existing method by correcting spelling mistakes, adding emoticons to the dictionary and exploiting some of the characteristics of online text, such as the use of exclamation marks. Much research has taken on the particularly difficult task of extracting sentiment from Twitter. Most of these approaches have relied on machine learning models that are sensitive to emoticons and the characteristics of tweets [46–49], although in some cases, linguistic structure was analysed [50].

Text messages and online commentary in video games have some unique characteristics. They are more conversational in style, as opposed to the more typical broadcast nature of tweets. They also tend to develop more community-internal norms, in particular with respect to abbreviations. Research from a linguistic or communication point of view has addressed the nature and characteristics of short messaging (e.g. [51]). To our knowledge, however, no sentiment analysis treatment of online commentary, and specifically online game commentary, exists.

In the present study we adapt a model originally designed for sentiment analysis of movie reviews to real time strategy video game chat. We also examine whether models designed for sentiment analysis can be ported to the detection of toxicity in competitive games. The shift to video game data is pronounced. Chat in fast-paced competitive games is, at least in the present case, dyadic, peripheral to the goal of the task (i.e. winning the match), and is produced under stress and severe time constraints. For example, Herring et al. [52] performed conversation analysis of a first-person shooter video game, and found that most messages are short in length. As such, we can expect chat in games to contain features such as short, abbreviated, and poorly spelled messages involving emoticons and game specific vocabulary. This presents a challenge to traditional lexicon-based and to machine learning models that have been designed for or trained on more conventional datasets such as movie reviews or consumer reviews.

Of course, game data contains sources of richness which go beyond differences in sentiment-rich vocabulary. The target of sentiment is presumably more ambiguous in game data. For example, the target of sentiment in a movie review is often explicitly stated

in the text (the plot, the actors, the characters, etc.), but players in the game need not make the target of the sentiment explicit in the text if the topic of their conversation is about an in-game interaction between the two players. Moreover, sentiment could be also directed towards an opponent, one's own performance, or the game itself.

Game data has conversational structure and contains domain-specific speech acts. For example, it is polite to type the greeting 'gg' (acronym for 'good game') at the end of the game, while it is rude for a winning player to type 'gg' to a losing player before the game has actually ended. Another obstacle to the detection of sentiment in games is the possibility that emoticons might serve many functional roles [53]. For our purposes, one important consideration is that emoticons, in addition to conveying sentiment directly, sometimes have the capacity to modify the sentiment of entire expressions. For instance, the writer of "fuck you:)" is swearing, but the emoticon mitigates the message and indicates that she swears in jest.

There are also special challenges in the automatic detection of toxicity. In an industry setting, a useful toxicity detector needs to keep hit rates and false alarm rates under strict control. When predicting toxicity, false alarms are especially grievous errors for two reasons. First, false alarms could lead to the incorrect labelling of individuals as toxic, and one might argue that false accusations are more problematic than missing toxicity, especially when toxic players might be revealed either by other methods or by application of a system so that players receive bans if the model detects some threshold number of toxic games. Secondly, and perhaps more importantly, controlling false alarm rates is important for managing the workloads of human raters. In many popular games, there are many thousands of games being played a day. With high volume, any system that does not tightly control false alarms is financially infeasible. For example, Valve, the company creating the real time strategy game Dota 2 releases data on the number of games played per day. They report about 600,000 games playing at 9:45 PST, March 6, 2017 [54]. If a detector had a false alarm rate of 1%, it would find about 60,000 false alarms over the course of a single hour.

A final challenge is that, because online multiplayer games match players according to skill, and to geographic region (to ensure that games are unimpeded by slow network speeds), players in multiplayer games could reasonably form relatively isolated sub-communities. It would be no surprise if overall rates of positivity and negativity, or our capacity to detect sentiment, varied by region of play.

1.4. Research questions

In investigating the applicability of a lexicon-based sentiment extraction model, The Semantic Orientation Calculator (SO-CAL), described below, to video game chat we perform three main analyses to investigate four important research questions:

- 1) *Emoticon-Blocking and Repetition Discounting Analysis*: We added two rules to SO-CAL that may, or may not, aid in the calculation of sentiment in the context of informal conversation. The first rule allows the sentiment value of text to be blocked by an emoticon that conflicts with the preceding text - as in "fuck you:)". The second rule discounts the sentiment of words as they are repeated - as in "that was a really really dirty move". SO-CAL has parameters that toggle the usage of the two rules. Thus, our question is: What is the contribution of these rules to overall model performance in the context of both sentiment classification (negative, neutral, positive), and in toxicity identification (toxic, non-toxic)?

- 2) *Dictionary Analysis*: What is the role of the various SO-CAL dictionaries (i.e., Profanity, Emoticons, Game Terminology, Greetings, Abbreviations) in the model's performance in both the sentiment classification task, and the toxicity detection task?
- 3) *Niche Analysis*: To what extent are there Niche effects, such as differences by skill level, by player region, by message length that might influence model ratings?
- 4) *Followup Toxicity Analysis*: Can we verify that our best toxicity detector retains its precision in a new sample?

Our first goal is to determine whether the Emoticon Blocking rule and the Repetition Discounting rule improve the accuracy of the model. This analysis is important not only for classifying sentiment in StarCraft 2, but also for probing the broader functions of emoticons and repetition in online chat messaging. Since emoticons and repetition are likely to be found in a wide range of online chat domains, this goal is of more general relevance to the analysis of sentiment. The Emoticon Blocking rule determines whether SO-CAL should treat emoticons in the same manner as regular words, with a sentiment score contributing to the whole, or as a marker which may alter the meaning of the entire sentence. When activated, emoticon-blocking replaces the sentiment score of entire sentences with the sentiment score of the emoticon. The Repetition Discounting rule discounts the value of each repetition of a word by half, such that the third use of the word has one third its initial value. We will extract sentiment from the text four times, once for each of the four combinations of the rules on, and off, and compare the accuracy of these models to see if these rules are a benefit.

Our second goal is to consider the impact of dictionary additions on model performance by comparing variations on our model's dictionary, since research is unclear as to how large this dictionary should be [33,55]. In this analysis, we will compare performance of the models that uses one of the new dictionaries, to a model that uses all the new dictionaries, and to a model that uses none of the new dictionaries. We expect this analysis to also provide a sense of how much work is required to port SO-CAL to new domains, as it may turn out that the dictionaries relevant to a particular classification problem are highly intuitive. For example, one might speculate that the profanity dictionary should be especially important for the toxicity detection. As with the Emoticon Blocking and Repetition Discounting Analysis, we perform the analysis separately for sentiment classification and toxicity detection.

Our third research goal is to evaluate human ratings and model performance across factors which are likely to impact model performance. Two of these reflect community or cultural differences, such as geographic region of play and player league. We see this analysis as a more direct test of SO-CAL's robustness to such differences. We also consider how human ratings and model performance depends upon the number of chat messages in each game. This message-length analysis is important because (a) the number of messages vary substantially from game to game and (b) prior studies have suggested that SO-CAL does not function as well when documents are short [33].

Given that we expect the number of extremely toxic games to be relatively small, our final research goal is to confirm the precision (the proportion of positive classifications that are correct) of our best toxicity detector in a distinct sample of 2025 games. These rates will be established by having humans rate games that are classified as toxic by our best toxicity detector, alongside an equally sized sample of games that are classified as non-toxic.

2. Methods

2.1. Dataset

StarCraft 2 (SC2) is a real-time strategy game in which players gather and manage resources to build an army and destroy their opponents. Each game begins with players possessing equal access to resources. Victory depends on a player's resource management, strategy, and control over their army. Any time in the game, players can open a chat box by pressing the 'enter' key on their keyboard, and send a text message. Pressing the 'tab' key allows the player to cycle through the list of recipients (individual players, allies, party, all), though we will only consider games with exactly two opposing players in the present work. After typing a message, pressing 'enter' again sends the message.

The data in this study, 3340 games in total, was originally collected by Thompson et al. [6]. For each game, a StarCraft 2 game replay parser, SC2Gears [56], outputs a text file that contains all the chat messages sent in the game, as well as one with global information about the game (such as players' tags, number of players, and so on) and another with in-game actions (such as directing units to move and attack), in a separate file. For each chat message, there is a timestamp, the sender's id, and recipient of the message. Participants were recruited through online StarCraft 2 community forums. Interested participants were referred to a survey asking about their prior gaming experience and gaming habits. At the end of the survey, they were prompted to submit one recent replay file of a game between the participant and another player. Importantly, our dataset only contains 29 females out of the original 3340 games, and so we cannot address gender differences (or generalize our results to women) in the present work.

Since chat messages for both players are included in the dataset, it was important that the conversations be anonymized. Player names were coded as individual numbers. Any identifying information that could threaten the anonymity of the players was removed, including names or gamer tags of the players or third parties, as well as place names that could reveal the location of the players. Of the 3340 replay files used by Thompson et al. [6], 314 games were excluded since they contained no chat messages, and one game was further excluded because of experimenter error. See Thompson et al. [6] for a complete description of the exclusion criteria that generated the original 3340 replay dataset. For our purposes, the most important of these criteria is the restriction of the present dataset to only one-versus-one games played online against random opponents chosen by the game engine (as opposed to custom games with friends). As a result, the recipient of the chat messages is always an opponent player. In total, the dataset for the present study contains 3025 games, and there are 15,489 messages (about five per game), 5244 unique word types and 34,060 word tokens.

The game itself has no character restrictions to message length, but instead is restricted to the amount of characters that can be displayed in the allotted space on the screen. Due to storage considerations and programmer error, our database can only store the first 100 characters per message, but only 5 out of 15,489 messages equal or exceed 100 characters. We partitioned all of our data into two sets. The initial dataset (henceforth referred to as the Primary Dataset) contains 1000 games with conversations to be manually rated for sentiment. Only one message within these games hit the limit of 100 characters or more, and this was accomplished through repeating the same character over fifty times (presumably by holding down a single key). The primary dataset is used to test our extension of SO-CAL directly. We reserve the remaining 2025 games for followup analyses (henceforth referred to as the Followup Dataset). Once we settle upon our best model for the detection of toxicity, we verify the model's precision by manu-

Table 1

Descriptive information for our primary dataset, which will be rated in its entirety by human raters, and the followup dataset. The latter is reserved for verifying our best toxicity detector.

	Primary dataset	Followup dataset
Number of messages	5046	10,443
Number of word tokens	10,742	23,318
Number of unique words	2446	4023

ally coding games in the followup dataset that the model classifies as toxic. Descriptives of the two datasets are presented in Table 1.

The present research was approved by the Office of Research Ethics at Simon Fraser University (Study number 2013s0019) as a secondary analysis of a dataset whose collection received earlier approval (Study number 2011s0302). Participants were 16 or older, and participants between the ages of 16 and 19 required parental permission to participate. Survey players provided informed consent (in the form of a checkbox) prior to taking an online survey and submitting a replay file. The present research also involved the analysis of anonymous chat data from opponent players who could not give informed consent. However, the games were played in an online setting in which confidentiality cannot be reasonably assumed. Authors did not have access to identifying information. The entire procedure, including the informed consent procedure and the analysis of all data sources, was approved by the Office of Research Ethics at Simon Fraser University (Study number 2011s0302;2013s0019).

All new data processing and analysis scripts for the present work underwent subversion control and was subjected to the following procedure. Data processing and analysis scripts were reviewed for logical integrity by an independent research assistant and their output manually verified by a third. Data ready for analysis was uploaded to MySQL tables which received final verification.

2.2. Human ratings of sentiment

Every game in the primary dataset was rated for sentiment on a 5-point Likert scale (toxic, mildly negative, neutral, mildly positive, strongly positive) by five human raters. All raters were members of the Cognitive Science Lab at Simon Fraser University, and two were authors. Three raters were experienced StarCraft players. The raters who were less experienced with StarCraft understood the basic mechanics of how chat is produced in StarCraft 2. In addition to the messages within the game, raters had access to message times, and were aware of the sender and recipient of each message.

Human rating began with a training session, where raters individually rated sentiment for 25 games from the primary dataset. Afterwards, raters discussed the rationale for ratings and resolved disagreements. After training, it became clear that a rule was required for rating games containing only polite greetings. Raters agreed that, unless politeness was offset by other sources of sentiment, games in which reciprocal greetings are present at the beginning and at the end of the game should receive a sentiment score of 1. Reciprocal greetings were defined as those containing two introductory greetings and at least one farewell greeting. We did not require two farewell greetings because final farewells can be cut off in the present dataset.

One week after training, raters independently rated all games. Following Siegel ([57], pp. 229–237), we evaluated inter-rater reliability for all five raters by computing Kendall's coefficient of concordance, with the adjustment for tied ranks (Kendall's $W=0.842$). Final human ratings were defined as the mode rating of the five raters. The vast majority of games (91%) were rated as neutral or mildly positive (see Table 2), and 2.7% of the games were classified

Table 2

Human ratings of sentiment.

Sentiment	Count (n = 1000)
–2 (Toxic)	27
–1 (Mildly negative)	38
0 (Neutral)	431
1 (Mildly positive)	485
2 (Strongly positive)	19

as toxic. With these human ratings in hand, we can test SO-CAL's capacity to predict sentiment in StarCraft.

2.3. SO-CAL modifications

The Semantic Orientation Calculator (SO-CAL) is a general-purpose system for sentiment analysis. It is a lexicon-based model, which has three advantages over a machine-learning model: (1) It is a lightweight alternative to computationally heavy machine learning sentiment extractors, (2) training is not necessary to adapt the model to a novel genre (though testing is usually required), and (3) the calculation of sentiment is theoretically transparent and informed by current linguistic theory.

Here, we briefly summarize the workings of the system, and how it was modified for StarCraft 2 chat conversations (For an in-depth description of SO-CAL, see [33]). SO-CAL contains five dictionaries for adjectives, nouns, verbs, adverbs, and intensifiers that bear sentiment. Sentiment scores for each word are rated on a scale from –5 to 5. Words that do not bear sentiment (have sentiment score of 0) are not stored in the dictionaries. SO-CAL scans the text for sentiment-bearing words and retrieves their sentiment scores.

After retrieving the sentiment scores of individual words, SO-CAL analyses the syntactic structure of the text and adjusts the scores if needed. The syntactic environment may necessitate enhancing or diminishing a token word's sentiment score: for example, intensifiers enhance the sentiment scores of the words that they modify, and negation would change the polarity of the sentiment of the phrase that it negates. In addition to negation and intensification, SO-CAL also takes into account *irrealis*, that is, words or phrases that convey a hypothetical concept and not a true opinion. For instance, SO-CAL ignores text that is in questions, because a question such as "Did you think this was a good game?" does not mean that someone thinks that the game was good nor bad, despite containing the word "good", which has positive sentiment. After the adjustment of sentiment scores based on syntactic context, SO-CAL averages the sentiment across the words in each sentence.

Finally, SO-CAL's parameters can be set so that additional weighting can be selectively applied when certain conditions are met. In the original SO-CAL, it was found that augmenting negative words by a weight of 1.5 improved model performance [33]. Previous versions of SO-CAL also have a parameter setting that allows for weighting sentences according to their position in the text, which is a simple method to implement some knowledge of discourse into the SO-CAL model, as in the online consumer reviews originally analysed, the end parts of the text were considered to be better predictors of overall text sentiment. The parameter for implementing weighting by position is turned off for the analysis of in-game chat data, since it has a different discourse structure than a consumer review, and there is no clear theoretical reason to motivate weighting by position. However, we do consider employing the repetition parameter, which reduces the sentiment of a repeated word (e.g. the third occurrence of a word will only contribute one third of its normal sentiment score).

The entire chat dialogue spanning one game is treated as one text by SO-CAL. Although game messages may consist of a phrase or even a single word or emoticon, SO-CAL considers each message to be a "sentence". That is, for the purposes for identifying syntactic dependencies between words to apply adjustments to sentiment score, the messages' start and end are boundaries in which syntactic relations can be found. If a message contains more than one sentence, then both the message boundaries and sentence boundaries apply.

Speaker information is not given to SO-CAL, so it is blind as to which messages belong to which speaker. This means that SO-CAL provides sentiment scores for entire games, not individual speakers. While SO-CAL could produce sentiment scores for each speaker, the current work focuses on rating the sentiment of the game as a whole.

In order to update SO-CAL's dictionary, a list of all distinct words and their frequency of occurrence was generated. Words were identified using words occurring more than once in the conversations in the primary dataset. Human coders would classify words as belonging to the profanity, terminology, emoticons, and greetings dictionary. Coders would then assign words a sentiment score and add them to the appropriate SO-CAL dictionary. We describe this process in more detail below.

One author and one research assistant manually compiled a list of words to add to the dictionary. The changes to the dictionary fall under the general categories of profanity, game terminology, abbreviations, vocatives, greetings, and emoticons. Most of these changes encompassed additions to the dictionary, but a small portion included changes to the sentiment of existing words. In such cases, the new game-specific meaning replaces the old sentiment score. For example, *fuck* had a rating of -4 in the SO-CAL dictionary before, and has a new SO of -3 .

The process of determining the sentiment score for the new game-specific words worked as follows: A team of three research assistants who were experienced with the game underwent a practice session to assign sentiment scores of the old dictionary items. One of the three raters of lexical sentiment was an author who was not involved in the rating of overall conversation sentiment. A second rater was involved in rating overall game sentiment, but this individual had no knowledge of SO-CAL's workings. Raters were trained on 44 sample words from the SO-CAL's original adverb, adjective, noun and verb dictionaries. Words were rated on an 11-point rating scale, the same scale used in SO-CAL's existing dictionaries. Raters were instructed to "Rate each word on a 11-point scale from most negative (-5) to most positive (5). Rate 0 for neutral." Afterwards, each rater compared their ratings with the sentiment scores of the words in the dictionaries, and discussed their choices.

After agreement was reached in training, the research assistants were asked to perform the same task to the new dictionary additions. The mode of the three ratings were taken as the word's final sentiment score. Words that are already present in the dictionary were given new ratings. Words that were absent in the dictionary previously were appended to the dictionary. Most of the unique words in the primary dataset were given a mode rating of 0 (neutral). SO-CAL treats any word that is not in its dictionary as having a sentiment score of 0, so words rated neutral were not appended to SO-CAL's dictionaries. If a word that was rated as neutral happened to be present in the original dictionary, it was removed. In the end, 152 new nouns were added to the noun dictionary, 23 new adjectives were added to the adjective dictionary, and 25 new verbs to the verb dictionary.

As shown in Table 3, the novel dictionary additions can be grouped into six categories (profanity, game-specific terminology, abbreviations, vocatives, greetings, and emoticons). Table 3 describes the changes to the ordinary lexical components of the

dictionary, and Table 4 describes the addition of emoticons. We speculated that the Profanity dictionary, containing words such as 'fuck' and 'shit', would be especially important in toxicity detection. We added dictionaries for Game Terminology, words and acronyms that are specific to the StarCraft 2 (e.g., 'cheeze', 'rush', and 'bm' (acronym for 'bad manner')), and Abbreviations, which include acronyms that are common to the overarching genre of computer mediated communication that videogame chat belongs to (e.g., 'ty' and 'afk', acronyms for 'thank you' and 'away from keyboard', respectively), in order to better capture the community under investigation. Since the messages under discussion involve multiple speakers, we also include a dictionary for Vocatives, which are terms that are used to directly address another member of the conversation (e.g., 'bro' and 'sir'), and Greetings, which are conversation openings or closings which may or may not be specific to a gaming context (e.g., 'hey', 'bye', 'glhf' (acronym for 'good luck have fun'), and 'wp' (acronym for 'well played')).

The final change to SO-CAL's dictionaries was the addition of emoticons (Table 4), which differed slightly from the processes of adding words to SO-CAL dictionaries. Although researchers tried to identify emoticons to the best of their abilities, it was not always clear whether a cluster of characters are an emoticon, a spelling mistake, or something else. To avoid misclassifying typographical errors as emoticons, a cluster of characters were only considered an emoticon if its frequency was greater than 1. Rather than rate emoticons on an 11-point scale, which may be too fine-grained, emoticons were manually sorted into 3 categories, depending on whether they expressed positive, neutral, or negative sentiment, and assigned a sentiment score of 1, 0, or -1 respectively. Of the 78 emoticons added to the dictionary, 29 were rated as positive (e.g. emoticons expressing happiness or love), 17 as neutral (e.g. emoticons expressing surprise), and 32 as negative (e.g. emoticons expressing confusion, anger, sadness, and annoyance).

Three modifications were made to SO-CAL to adapt it for parsing emoticons. The first was implementing case-sensitivity towards emoticons. SO-CAL is case-insensitive towards words, but an exception has to be made for emoticons. While some emoticons have equivalent uppercase or lowercase forms, such as 'XD' and 'xd', other emoticons do not. One example is 'D:', which is negative, and 'd:', which is positive. We modified SO-CAL to recognize such distinctions.

The second modification was to disable valence shifting for emoticons. The non-linguistic nature of emoticons means that they do not have a clear syntactic role. Oftentimes emoticons can stand alone and whole messages can contain an emoticon and nothing else. Otherwise, emoticons are usually tagged onto the ends of messages. Therefore, SO-CAL was modified to make emoticons exempt from syntactic processes that alter the sentiment score of a word, like negation or intensification. Furthermore, textual communication lacks intonation, which may contain cues for emotion or sarcasm. It seems plausible that emoticons are used to mitigate the strength of a sentence, or to avoid face-threatening situations by changing the tone of the sentence, or to signal sarcasm. In order to account for this, we created a new parameter for SO-CAL, emoticon blocking. Similar to the implementation of intensification in the model, emoticon blocking gives an emoticon the ability to change the sentiment of the words that precede it in the same message. The sentiment score of the other words in the sentence were, in such cases, set to 0, and the average sentiment score for that sentence was simply the emoticon's rating. In the example message "Well gg, I suck at terran:)", the sentiment score for "Well gg, I suck at terran" is ignored.

The last change involved having the model sum sentiment scores across sentences instead of averaging, in an attempt to better account for conversational structure. Conversations have socially acceptable conversation openings and closings; within the

Table 3

Additions to the dictionaries, by category. Additions include both new words and changes to the sentiment of old words.

	Nouns (n = 187)	Adjectives (n = 42)	Verbs (n = 43)	Adverbs	Intensifiers
Profanity	52	14	17		
Gaming/game-specific Terminology	50	26	26		
Abbreviations	14	2			
Vocatives	5				
Greetings	66				
Dictionary size					
Original dictionary (unique counts)	1539	2819	1130	878	216
New dictionary	187 changes: 11 changed SO + 152 additions – 24 deletions = 1667	42 changes: 8 changed SO + 23 additions – 11 deletions = 2831	43 changes: 3 changed SO + 25 additions – 15 deletions = 1140	878	216

Table 4

Frequency of positive, neutral, and negative emoticons, with examples from each category.

Emoticons (n = 78)	Count	Examples
Positive	29	^^ XD ;) <3
Neutral	17	:O >_> o.O 0-0
Negative	32	D: :(-.- T_T :S

context of gaming, gamers will greet opponents with 'glhf' and end the game with 'gg' or 'wp'. Whether players go through these conventions is related to the overall sentiment of the conversation. An averaging system would mask the difference between a conversation where there is a greeting and reply and another conversation where there is a greeting and the greeting is not reciprocated.

3. Results

This section is divided into four subsections, one for each of the four main analyses: the Emoticon Blocking and Repetition Discounting Analysis, the Dictionary Analysis, the Niche Analysis, and the follow-up Toxicity analysis. Confusion Matrices for all classifications can be found in Supplementary Materials.

3.1. Emoticon blocking and repetition discounting analysis

Our first research question required testing the efficacy of emoticon blocking and repetition discounting in sentiment classification. To test this, we ran SO-CAL four times on the complete dictionary, once for each of the possible settings for emoticon blocking and the repetition discounting rules. We then use these initial sentiment scores to classify games into their sentiment category. Optimizing each version of SO-CAL for either sentiment or toxicity detection is a matter of comparing model fit across all possible semantic category cutoffs. In sentiment classification there were two cutoffs that distinguished the three sentiment categories (negative - *lower cutoff* - neutral - *upper cutoff* - positive), and in toxicity detection, there was only one (toxic - *cutoff* - nontoxic). In the case of sentiment, fit was defined in terms of overall accuracy. In the detection of toxicity we define fit as the difference between the true positive rate and a weighted false alarm rate. In order to account for the intuition that false alarms are the more grievous error, we chose to weight the false alarm rate by twenty times. The importance of false alarms depends on the needs of the user, and other weighting might be preferred in other situations. Because the cutoffs are determined after the model generates the sentiment score, this step is easily computed, and easily changed.

As can be seen in Table 5, all four models had accuracy over 80%, which is in keeping with the accuracies of many sentiment detection systems [20–24]. A similar analysis was done for the toxicity detection task, and the results can be seen in Table 6. As in the sentiment analysis, manipulating emoticon blocking and

Table 5

Emoticon blocking and repetition discounting analysis for sentiment classification.

	Lower cutoff	Upper cutoff	Accuracy
Emoticon blocking off Repetitions off	–4.50	2.25	0.80
Emoticon blocking off Repetitions on	–3.75	2.25	0.80
Emoticon blocking on Repetitions off	–4.50	2.25	0.81
Emoticon blocking on Repetitions on	–4.50	2.25	0.81

repetition parameters had relatively little influence on model performance. As with sentiment classification, the model where the Emoticon Blocking parameter is turned on and the Repetitions Discounting parameter is turned off (hereby abbreviated as the NoRepBlock model) appeared to enjoy a slight advantage according to our fit criterion.

The best fitting toxicity detection model has a hit rate far below that found in the sentiment classification task. In light of the fact that performance in the toxicity detection task must be evaluated based on both maximizing accuracy and minimizing false alarms, the best fitting model is simply one among many possible hit/false alarm ratios that the best fitting model can make. For reasons explained in the introduction, we took the minimization of false alarms to be more important than maximizing hit rate. To reflect this, our fit function valued false alarms as 20 times more important. In practice, this weighting can be changed after the sentiment is extracted to reflect the needs of the user. The cutoff value between toxic and non-toxic games also requires balancing hit rate with false alarm rate. We plotted hit rate and false alarm rate for the best fitting model at every possible classification boundary in an ROC-like curve (Fig. 1) to make this tradeoff clear. An examination of the curves reveals no substantial advantage of NoRepBlock over other complete-dictionary models. Further, it deserves emphasis that, while we prioritize the minimization of false alarms to the maximization of hit rate, these models could easily be optimized more heavily towards the latter.

While it is clear that the Repetition Parameter and Emote Block parameters do not greatly influence the models' performance in either the sentiment classification or the toxicity detection task, there are several possible reasons for this. First, it might be the case that there simply are not enough emoticons or repeated words in the dataset for these rules to matter much. Second, even if the rules are activated often, the changes in total sentiment score may be too small to cross a cutoff and thus change model predictions. Consequently, it makes sense to ask whether parameters at least drag raw sentiment score in the direction of accurate classification. Third, it is possible that even if classifications do change, that the rule might influence the classifications in both positive and negative ways, leading to little overall change in performance.

Table 6
Emoticon blocking and repetition discounting analysis for toxicity detection.

	Lower cutoff	Hit	False alarm (FA)	Fit value ((Hits% – 20*FA%))
Emoticon blocking off Repetition off	–12.00	0.37	0.009	18.5
Emoticon blocking off Repetition on	–9.50	0.41	0.011	18.1
Emoticon blocking on Repetition off	–15.00	0.33	0.007	18.9
Emoticon blocking on Repetition on	–11.00	0.37	0.009	18.5

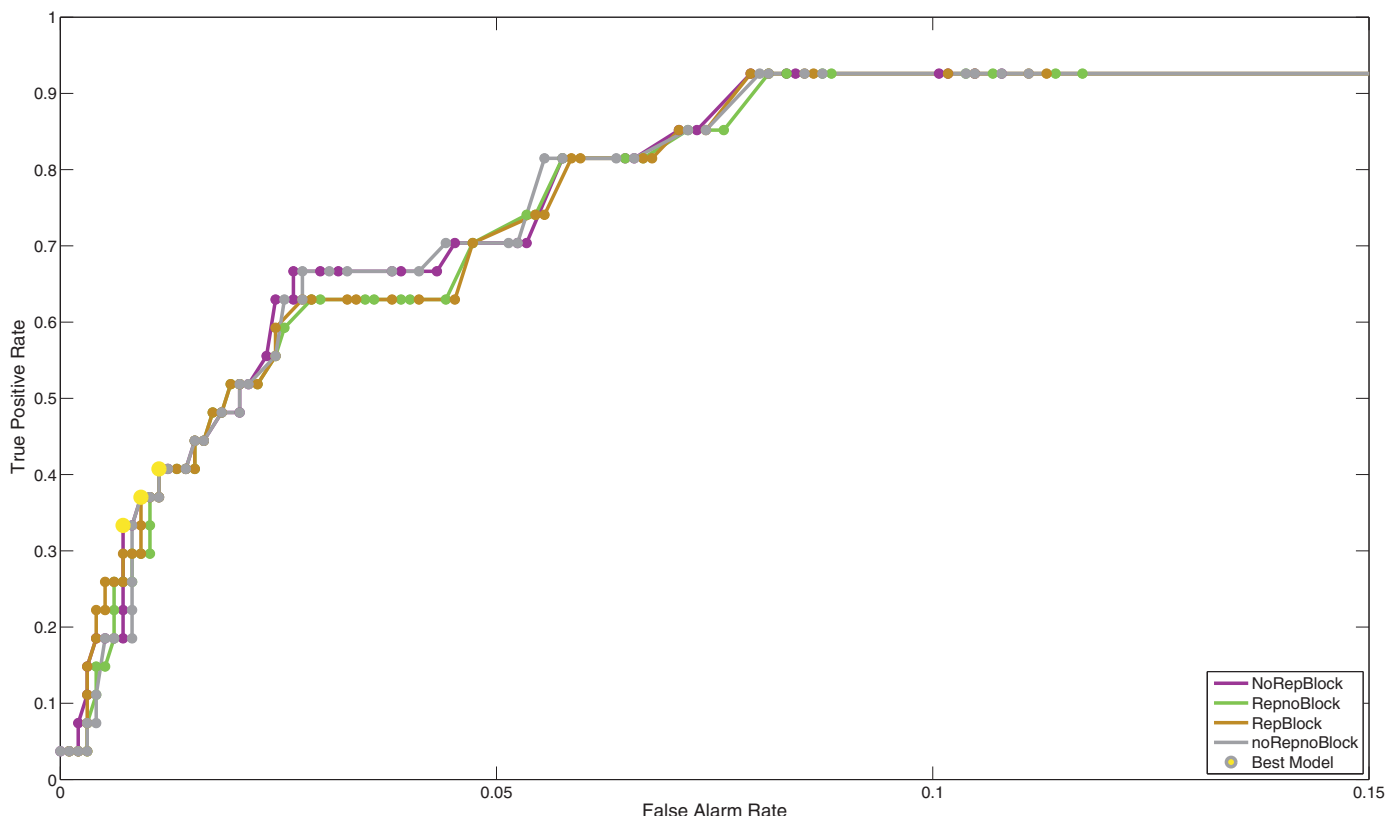


Fig. 1. Hit and false alarm rate tradeoff for all possible cutoffs in the toxicity detection task.

Investigating these possibilities for the repetition parameter first, we found that even prior to the selection of cutoffs, models with the repetition parameter on (RepNoBlock) differed in sentiment score from models with the repetition parameter off (NoRepNoBlock) about 51% of the time, suggesting that these parameters have ample opportunity to impact model output. Once the best possible cutoffs were selected, however, these differences in raw sentiment score rarely resulted in different model predictions (27 games). The models with the repetition parameter produced different predictions even more rarely in the case of toxicity (5 games). Furthermore, the repetition parameter was not always dragging predictions in the right direction. Of the 185 cases where both NoRepNoBlock and RepNoBlock were inaccurate in the classification of sentiment, RepNoBlock produced output which was closer to the relevant cutoff 36% of the time, and farther from the cutoff 21% of the time. Of the 24 cases where both models were inaccurate toxicity detectors, 79% of the time RepNoBlock was closer to the best cutoff and 21% of the time it was farther. Finally, when classifications change and only one model is correct, RepNoBlock was correct in 12 out of 26 games (2 out of 3 games for toxicity). Overall then, while it looks like the repetition parameter is often

dragging raw sentiment score in the right direction, its usefulness appears modest at best.

For the emoticon blocking parameter, the raw sentiment score of NoRepBlock and NoRepNoBlock differed in only 44 games, indicating that no matter how useful emoticon blocking was, it could not have much influence over model performance as a whole. Furthermore, though emoticon blocking was on in our best model, its sentiment classifications differed from NoRepNoBlock in only 10 cases (its toxicity predictions differed in 3 cases). Of the 189 cases where both models were inaccurate sentiment classifiers, NoRepBlock was closer to the cutoff 3.7% of the time and farther from the cutoff 2.7% of the time. In the 24 cases where both models were inaccurate in toxicity detection, NoRepBlock was closer to the cutoff 29% of the time and farther 71% of the time. Finally, in the 10 cases where only one model was correct, NoRepBlock was correct 8 times. It was correct 2 out of 3 times in the case of toxicity detection. Overall, emoticon blocking appears promising but the present data do not contain enough messages ending with emoticons for us to properly evaluate its utility.

Given that the distribution of sentiment is extremely unbalanced, we used a chi-squared goodness of fit test to determine whether NoRepBlock's accuracy during sentiment classification was

Table 7

Emoticon blocking and repetition discounting analysis for sentiment classification.

	Lower cutoff	Upper cutoff	Accuracy
Profanity	−4.50	0	0.507
Emoticon	−10.50	0.25	0.506
Game terminology	−8.25	0	0.503
Greetings	−6.50	2.25	0.780
Abbreviations	−5.00	0	0.501
None	−10.50	0	0.494

significantly higher than a classifier which always predicts the most common class (predicting that every game had positive sentiment would yield a classifier with 50.4% accuracy). Our best model, NoRepBlock, performed above baseline in sentiment analysis ($\chi^2 = 372.12$, $df = 1$, $p < 2.2e-16$). In order to get a better sense of how the model would perform in a balanced sample, we resampled 19 games from each level of human ratings (ranging from −2 to 2) without replacement, and then computed the trinary classification of sentiment (negative, neutral, and positive) using our most successful model, NoRepBlock. Across 1000 of these resamples, NoRepBlock accuracy was normally distributed with a mean of 70% ($sd = 4\%$; see Supplementary Fig. 1). To provide a balanced picture of the toxicity analysis, we sampled 27 toxic games with 27 non-toxic games without replacement and used NoRepBlock for toxicity detection. A total of 187 resamples out of 1000 had exactly one false alarm, and 19 had more than one (see Supplementary Fig. 2).

In summary, we find the performance of the model in both the sentiment classification task and the toxicity task to be excellent. In both tasks, turning emoticon blocking on and taking the sentiment of the emoticon to represent the sentiment of the utterance preceding it, and the repetition parameter off, meaning the model allows repeated words to carry their full sentiment, led to the best performance; we hasten to add, though, that differences between models were slight because emoticon blocking was rarely activated in our dataset.

3.2. Dictionary analysis

The Emoticon Blocking and Repetition Discounting Analysis performed in Section 3.1 identified the best settings for our model parameters, and these remained fixed for our remaining analyses. We then turned to the question of how influential each of the new Dictionaries are. Recall that we added entries to the dictionary of words for which we had sentiment ratings words from five different categories: Profanity, Emoticons, Game Terminology, Greetings, and Abbreviations (there were too few vocatives added to the dictionary for them to be included in the Dictionary Analysis). In this analysis, we compared the full model (with the complete dictionary and best parameter settings) with five new versions, each of which included the original SO-CAL dictionary plus one of the five new Dictionaries built for this project. We also included a version of SO-CAL with none of the new dictionaries. This repeat this analysis for sentiment classification and toxicity detection.

Table 7 shows the model accuracy for each dictionary. We found that the greetings dictionary had the most improvement over the model with no new dictionaries and, at 78% accuracy, it was nearly as good as the full model. The remaining dictionaries displayed unequivocally poor performance. The second best dictionary, Profanity, correctly classified only 13 more games than the model with no new dictionaries, and was only 0.3% more accurate than a model choosing the dominant class (positive sentiment) on every classification. Given that the vast majority of games in our database were rated as containing either neutral or mildly posi-

tive sentiment, it should be no surprise that the most important dictionary would be the one that distinguishes warm and polite conversation from games with little or ambiguous sentiment.

Table 8 describes the Dictionary Analysis for the toxicity detection task. The dictionaries most relevant to the detection of toxicity seemed to differ substantially from those in sentiment classification. Profanity and game specific terminology dictionaries had the most impressive overall fit values, though these were substantially lower than the fit values for the overall model.

As in the Emoticon Blocking and Repetition Discounting Analysis, looking at the ROC-like curves (Fig. 2) will give a better sense of the tradeoffs for each model. Of the dictionaries, only Profanity had a competitive curve, though the complete model showed superior performance at most points. The other dictionaries were clearly inferior, showing higher false alarms for a given hit rate, and a lower hit rates for a given false alarm rate.

In summary, we find that the inclusion or exclusion of specific dictionaries impacts performance far more than the emoticon blocking and repetition parameters. In the sentiment classification task, the Greetings dictionary was the strongest independent contributor, and for toxicity detection, the Profanity dictionary proved most useful.

3.3. Niche analysis

Generality of findings is always an important consideration. Because StarCraft 2 players play on servers specific to geographic region, and because leagues themselves are somewhat isolated communities (e.g. players in the highest league are very rarely matched against players in the lowest league), we wanted to examine whether human ratings of sentiment and, more importantly, the accuracy of our best model varied by region and league. For example, it might be that the additions to our dictionary might be culture-specific, boosting model performance in some regions but not others. Finally, given that there was a great deal of variability in the number of messages per game (Fig. 3A), we were interested in verifying that the predictive power of the model was not restricted to conversations of a particular length.

The differences in human ratings of sentiment and model accuracy varied little by region (Table 9), and NoRepBlock's accuracy on European and American servers differed only by about 1% (we lacked a sufficient sample size to consider the servers in Asia or South-Eastern Asia). Human ratings did seem to decline by about 0.2 from Bronze to Master leagues (Table 10), perhaps suggesting less congeniality as players become more competitive. Interestingly, 5 of NoRepBlock's 7 false alarms came from the lowest four leagues (57% of our data), resulting in the markedly different Fit values reported in Table 10. Most important, though, is that SO-CAL accuracy generally remained consistent across these leagues, with sentiment classification accuracy between 80–82%.

Classifier accuracy was more dramatically influenced by conversation length (see Figs. 3 and 4). NoRepBlock had an accuracy of 85% on the 706 games containing only four chat messages or less, and only 71% on the 294 games with 5 messages or more. A general picture of the relationship between conversation length and accuracy can be seen in Figs. 3 and 4.

Overall, then, the human ratings of sentiment were consistent across Regions and Leagues, as was model performance. Model performance was slightly better for shorter messages than for longer conversations, but model performance was good in all cases.

3.4. Follow-up toxicity analysis

In toxicity detection it is an important goal to minimize false alarms, and one might be willing to accept some missed toxic

Table 8
Dictionary analysis for toxicity detection.

	Cutoff	Hit	False alarm (FA)	Fit value (Hits% – 20*FA%)
Profanity	–25.25	0.111	0.002	7
Emoticon	–21.25	0.037	0	3.7
Game terminology	–11.50	0.148	0.005	4.5
Greetings	–28.00	0	0	0
Abbreviations	–28.00	0	0.001	–2.1
None	–27.25	0.037	0	3.7

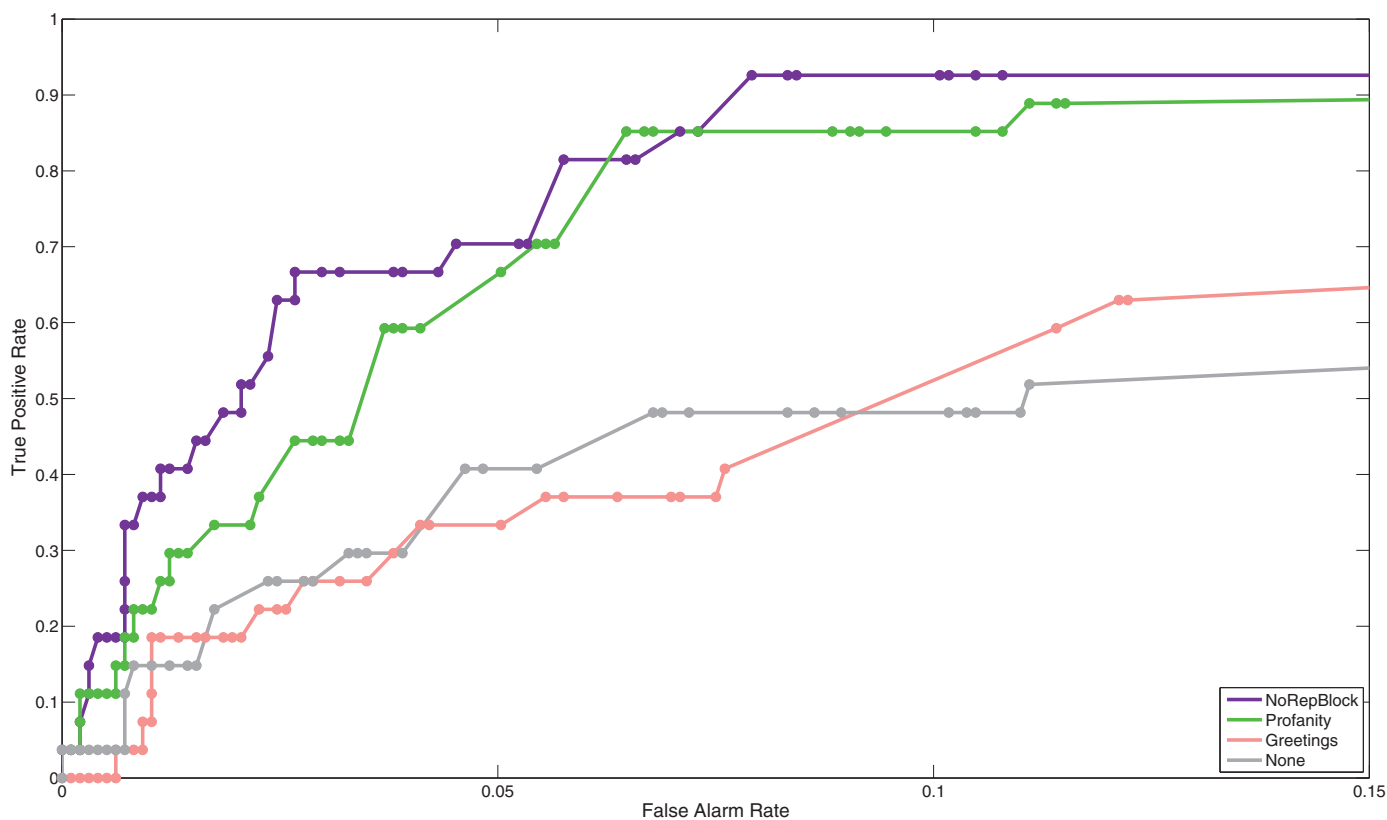


Fig. 2. Hit and false alarm rate for all possible cutoffs in toxicity detection (by dictionary).

Table 9

Ratings by Region. NoRepBlock cutoffs for sentiment classification and toxicity detection are reported in Tables 5 and 6 respectively. Fit values are undefined when there are no toxic games in the dataset and are otherwise equal to Hits% – 20*FA%.

Region	Count (n = 1000)	Mean of mode human rating (–2 to 2)	Mean NoRepBlock rating (continuous)	Accuracy (sentiment)	Fit (toxicity)
America	603	0.44	2.17	81	16.5
Europe	363	0.39	1.94	80	18
South-East Asia	33	0.64	3.38	82	NA
Asia	1	0	0	1	NA

Table 10

Ratings by League. NoRepBlock cutoffs for sentiment classification and toxicity detection are reported in Tables 5 and 6 respectively. Fit values for the toxicity analysis are undefined when there are no toxic games in the dataset. Leagues are collapsed in reporting Fit in order to ensure the presence of some toxic games.

League	Count (n = 1000)	Mean of mode human rating (–2 to 2)	Mean NoRepBlock (continuous)	Accuracy (sentiment classification)	Fit (toxicity detection)
Bronze	42	0.50	2.01	0.81	0.15
Silver	106	0.59	2.61	0.81	
Gold	175	0.46	2.63	0.82	
Platinum	246	0.47	2.13	0.80	
Diamond	237	0.37	1.54	0.81	0.24
Master	185	0.32	2.03	0.81	
GrandMaster	9	0.33	1.48	0.89	NA

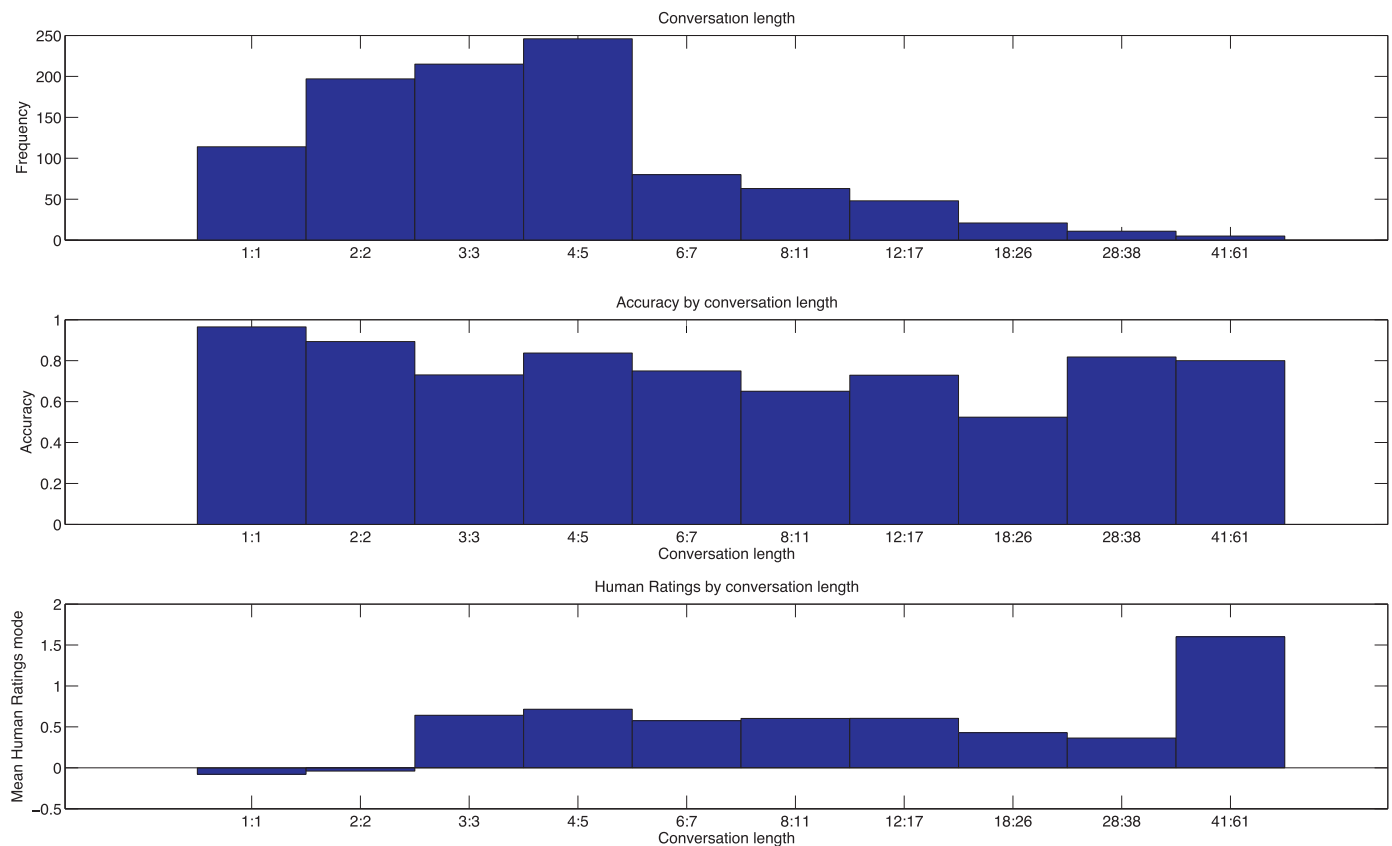


Fig. 3. Descriptives for sentiment analysis. (A) Histogram of log conversation length (the number of messages in a game), (B) Accuracy by conversation length, and (C) Human Ratings by conversation length for the sentiment analysis. Bins are of equal size in units of log conversation length.

games in order to avoid punishing a player or community member for toxic behaviour unjustly. In a follow-up analysis we sought to verify that our model keeps false alarms under control even when ported to a new dataset. We ran NoRepBlock, using our best parameter settings for toxicity detection, on an additional 2025 games from the SkillCraft dataset [6]. In our original dataset, this model had 9 true positives and 7 false alarms, so we expected our toxicity detector's precision - the proportion of positives that were true - to be around 56.2%. Furthermore, given NoRepBlock's false alarm rate of 0.7% and hit rate of 33% in the original dataset (also given the assumption that the base rates of toxicity are about 2.7% for both datasets) we would expect our model to produce about 14 false alarms and correctly identify about 18 true positives.

Our model classified 21 games as toxic, and 2004 as nontoxic. We then had two of the original raters classify the 21 games the model indicated were toxic games, and an additional 21 randomly selected non-toxic games, to allow us to calculate model precision. Of the 21 games the model identified as toxic, 10 were judged to be toxic by human raters, and 11 were judged to be false alarms, a precision of 47.6%, though two of these false alarms were based on games where the raters themselves disagreed about the games toxicity. We further identified one miss, a toxic game that the model had classified as non-toxic. Overall, our results are in keeping with our general finding that SO-CAL can be tuned to detect toxicity while keeping the number of false alarms under control.

4. Discussion

The present research shows that a lexicon-based approach is a viable method for detecting sentiment in video game environments. The best fitting models had accuracy that was well above

baseline for the detection of sentiment, and our toxicity detector was capable of keeping the number of false alarms under control in both the Primary and the Follow-up dataset. These results are comparable to those in other sentiment analysis systems [3,49,58,59], where it seems that 80% accuracy is relatively common, although improvement over that proves to be difficult and is very much domain-dependent [20]. As in prior research [36], the tailoring of SO-CAL's dictionary to a new domain was relatively easy, and the Dictionary Analysis revealed that these additions are much more important to overall model performance than the emoticon blocking and repetition parameters. Finally, the Niche Analysis revealed that model performance was relatively stable across region, league, and message length. The fact that we were able to take a model which was designed to extract sentiment from movies reviews and adapt it, in a straightforward manner, to the radically different domain of in-game chat within a video game, seems to align our results with other work suggesting that lexicon-based approaches can be robust across domains [33].

A major purpose of the emoticon blocking and repetition discounting analysis was to identify whether these rules offer the best performance and, though we achieved slightly better model performance when turning repetition off and turning emoticon blocking on, it appears that performance of the model varied little with different parameter settings. Closer inspection revealed no evidence that the repetition parameter was helpful, and emoticon blocking was activated too rarely in the dataset for us to properly evaluate its utility. Therefore, while existing research does suggest that emoticons can be used to indicate sarcasm in Twitter messages [60], the utility of emoticon blocking in extracting sentiment from StarCraft 2 chat remains unclear.

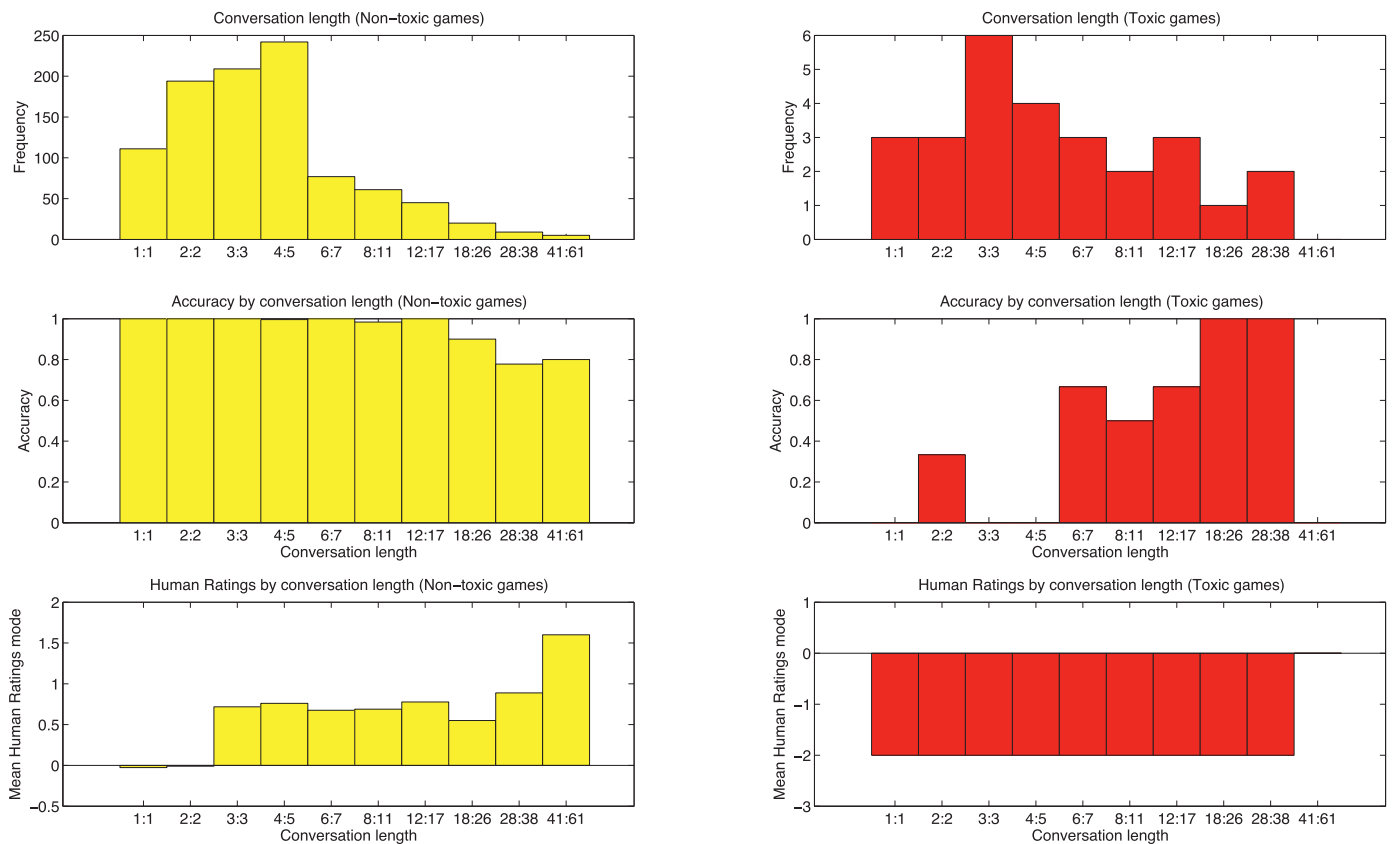


Fig. 4. Descriptives for toxicity analysis. Frequency, Accuracy, and Human Ratings by conversation length (the number of messages in a game) where games are classified as non-toxic (yellow) and toxic (red). Toxic games are those with a human rating of -2 . Bins are of equal size in units of log conversation length. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The Dictionary Analysis suggests that dictionaries must be tailored to the classification task at hand. Profanity use is helpful for the task of identifying strongly negative conversations, but it confers only a small benefit in the general task of sentiment extraction. Conversely, greetings are absolutely essential for general sentiment analysis. All other dictionary sets produced models which were only 50.7% accurate or less, and these were only a little better than a baseline model which predicts that every game has positive sentiment (50.4%). Sentiment detection, in a domain such as StarCraft 2, appears to be largely a matter of tracking politeness, and so it is plausible that future work in this domain should consider adopting coding conventions that distinguish sentiment and sentiment due to politeness.

The Niche Analysis revealed that, outside of regions and leagues with small sample sizes, model performance did not vary much by Niche (2% or less). It is important to note that while accuracy varied more by message length, overall accuracy seemed to remain high (above 75% in our sample). This is especially important in light of the finding that the greetings dictionary is essential to the overall model's performance, and since human raters settled on a convention to rate games with reciprocal greetings as a 1 (but no other sentiment to speak of). If our results were entirely due to this decision, we would expect model accuracy to drop off more sharply in games with higher conversation length. After all, producing reciprocal greetings requires three lines of conversation (two introductory greetings and at least one farewell greeting), and so games with higher message counts should be more weakly impacted by this rule.

The Dictionary Analysis, Niche Analysis, and followup analysis all speak to the effectiveness of the lexicon-based approach. The Niche Analysis revealed that our best model, supplemented with

the complete dictionary, performed well across region, league, and message length. Only message length appeared to have any impact on model performance, but accuracy remained above the accuracy which would result from always predicting the most prevalent class. The creation of the dictionaries were relatively straightforward, and the Dictionary Analysis revealed that the importance of dictionary was highly intuitive. In the classification of sentiment, where the vast majority of the games were rated as containing neutral or mildly positive sentiment, domain-specific vocabulary relating to greetings and politeness seemed very important. In the detection of toxicity, domain-specific profanity was much more useful. This is fortunate, as it might have been the case that the importance of dictionaries to model performance was so counter-intuitive as to have necessitated the use of machine learning and large training sets in their creation. Finally, while the precision of our toxicity detector changed from 56.2% to 47.6% when the model was applied to an additional 2025 games, perhaps due to slightly different rates of toxicity in the two datasets, the classifications of our model did not produce an exorbitant number of false alarms. We take this to be very important in applied settings, where accusations of toxicity are too serious not to be followed up on by human ratings. More generally, our results are in keeping with previous findings about the versatility of lexicon based approaches [33,61,62].

It appears plausible that the model might be extended to the detection of cyber-bullying, as the overt aggression in toxic games (See Supplementary Materials for examples) is presumably sufficient for popular definitions of internet harassment [e.g. 63]. Of course, toxicity in StarCraft 2 does not meet definitions of cyber-bullying which require repeated attempts to harm others [e.g. 64], as players part ways after a fifteen minute game. Nevertheless, re-

peated harassment would presumably be even easier for models to detect, suggesting that the prospects for detecting some forms cyber-bullying in online contexts are good.

One final thing to note is that although we treated toxicity detection and sentiment classification as separate tasks for implementation and expository purposes, in actual use it is possible for a single model to do both tasks. Only the cutoffs differed between our best sentiment classifier and toxicity detector. It is possible, therefore, to assign cutoffs for different goals, and examine both toxicity and commendable friendliness with a single run of the model.

4.1. Limitations

The research described here has several limitations. First, the present work looks at the identification of sentiment in video game chat messages using data from only a single game, StarCraft 2. This game contrasts in many ways (length of messages, cognitive constraints on the participants, etc.) to the kinds of data found in the movie reviews to which SO-CAL has thus far been applied, and so provides a good test case of its versatility. Nevertheless, it is impossible to know, without further study, if the version of SO-CAL that was used here on StarCraft 2 chat would generalize to other games. It is our impression, based on discussions with people that play many video games, that lexicon items such as 'glhf', 'afk', and 'gg', are in widespread use amongst people who play online games, and so we are optimistic that SO-CAL may do reasonably well; it remains an empirical question.

Second, the data used in the present analysis was submitted by the players themselves, and so may underrepresent the level of toxicity (which was low) amongst StarCraft players. Given that our purpose was to test SO-CAL's ability to extract sentiment in this domain rather than assess the level of toxicity itself, it does not likely impact the conclusions of our study. Nevertheless, while it is true that toxicity is rather rare in our sample, its plausible that toxicity may be contagious, implying toxicity could grow over time if left unchecked. Previous studies of toxicity in gaming (e.g. [14]), point out that toxicity may be especially problematic in driving away new players. Unfortunately, the proportion of toxic games in our sample does not reveal anything about the number of players who may have quit on account of experiencing toxicity. In our sample, 3043 out of 3395 players report over 100 h of StarCraft 2 experience (roughly 400 games), suggesting that toxicity is likely to have been experienced by the majority of our sample.

Finally, we used the mode of the scores of human raters, rather than the average, to assess the performance of the model. The fact that human raters can disagree, not just on the sentiment, but on toxicity as well, suggests that the task itself is not straightforward, and that perfect performance is not possible. Looking at the strengths and weaknesses of different methods for arriving at human estimates of sentiment can therefore be of some value for future research.

4.2. Conclusion

Our general finding is that a lexicon-based sentiment analysis approach designed for movie reviews can be extended to the detection of sentiment and toxicity in a video game with modest effort and no additional training data. Labelled data is still required, of course, for testing the extension of SO-CAL to new domains, but the size of the dataset is substantially smaller than what would be required for a (supervised) machine learning approach. This feature makes lexicon-based sentiment analysis an appealing option for both academic and industry use.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.knosys.2017.09.022](https://doi.org/10.1016/j.knosys.2017.09.022).

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